Jan 2020 Sample Solutions.

Q1 a) $A = \begin{bmatrix} 1 & -3 \\ 0 & 2 \\ 4 & -2 \end{bmatrix}$ $B = \begin{bmatrix} 2 & -2 & 0 \\ 8 & 1 & 1 \end{bmatrix}$

i) size = Rows x Cols

$A = 3 \times 2$, $B = 2 \times 3$

ii) $a_{22} = 2$, $b_{23} = 1$

iii) Transpose of $B \Rightarrow B^T = \begin{bmatrix} 2 & 8 \\ -2 & 1 \\ 0 & 1 \end{bmatrix}$

iv) $2(A + B^T) = 2\left(\begin{bmatrix} 1 & -3 \\ 0 & 2 \\ 4 & -2 \end{bmatrix} + \begin{bmatrix} 2 & 8 \\ -2 & 1 \\ 0 & 1 \end{bmatrix}\right)$

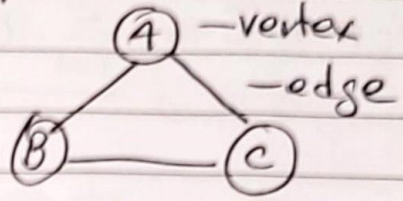
$= 2 \begin{bmatrix} 1+2 & -3+8 \\ 0+-2 & 2+1 \\ 4+0 & -2+1 \end{bmatrix}$ add the corresponding elements

$= 2 \begin{bmatrix} 3 & 5 \\ -2 & 3 \\ 4 & -1 \end{bmatrix}$

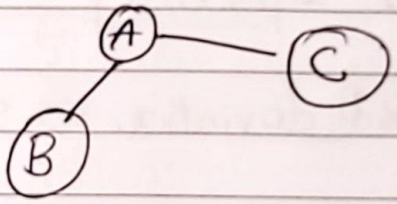
$= \begin{bmatrix} 6 & 10 \\ -4 & 6 \\ 8 & -2 \end{bmatrix}$

b) See Notes:

Graph: consists of vertices & edges such that every edge is attached at each end to a vertex
+ example graph



Tree: connected graph with no cycles, if there are n vertices then there are n-1 edges...
+ example Tree



3 vertices
3-1/(2) edges.

c) $t(1) = 3$
$t(n) = t(n-1) + 4 \quad (n>1)$

$t(2) = t(2-1) + 4$
$= t(1) + 4$
$= 3 + 4 = 7$

$t(3) = t(3-1) + 4$
$= t(2) + 4$
$= 7 + 4 = 11$

$t(4) = t(4-1) + 4$
$= t(3) + 4$
$= 11 + 4 = 15$

d) See Notes ....

e) 2, 12, 17, 19, 25, 32

i) Mean $\Rightarrow \bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$

$$= \frac{2+12+17+19+25+32}{6}$$

$$= \frac{107}{6}$$

$$\bar{x} = \boxed{17.83}$$

ii) Standard deviation $\Rightarrow s^2 = \frac{\sum x_i^2 - n(\bar{x})^2}{n-1}$

$$s^2 = \frac{2^2+12^2+17^2+19^2+25^2+32^2 - 6(17.83)^2}{6-1}$$

$$= \frac{4+144+289+361+625+1024-1907.4534}{5}$$

$$= \frac{2447-1907.4534}{5}$$

$$s^2 = 107.91$$

$$s = \sqrt{107.91} = \boxed{10.38}$$

f) i) $P[8 \text{ and a } 2] = 2/36 = 1/18$

↑ 2 Dice & total possible outcomes $= 6 \times 6 = 36$

ii. $P[8 \text{ and a } 2]$ (Dice 1, Dice 2) or $P[2 \text{ and a } 3]$ (Dice 1, Dice 2)

$\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$ or $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$

$\Rightarrow \frac{1}{36} + \frac{1}{36} = \frac{2}{36}$

ii $P[\text{PRODUCT} >= 3]$ Build event space for Product

X	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	8	10	12
3	3	6	9	12	15	18
4	4	8	12	16	20	24
5	5	10	15	20	25	30
6	6	12	18	24	30	36

$P[\text{Product} >= 3] = 33/36 = 11/12$

g) $f(x) = 3x + 5$

$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$f(x+h) = 3(x+h) + 5$

$= 3x + 3h + 5$

$= \lim_{h \to 0} \frac{3x + 3h + 5 - 3x - 5}{h}$ (A, B)

$\frac{dy}{dx} = \lim_{h \to 0} \frac{3h}{h} = \boxed{3}$

$\therefore \alpha = 3$ and $\beta = 3$

h) $\int_1^3 (2x^2 + 3x + 1)\, dx$

$= \frac{2x^3}{3} + \frac{3x^2}{2} + x \Big]_1^3 = \left(\frac{2(3)^3}{3} + \frac{3(3)^2}{2} + 3\right) - \left(\frac{2(1)^3}{3} + \frac{3(1)^2}{2} + 1\right)$

$= \boxed{\cancel{27.83}\ 31.33 \text{ square units}}$